

RETAIL SALES PERFORMANCE ANALYSIS

Business Problem Statement & Project Documentation

Storyline: Growth | Margin Quality | Operational Efficiency

Analyst	Domain	Tools Used	Report Type	Pages
Yogesh Gariya	Retail / E-Commerce	MySQL · Power BI · DAX	End-to-End BI Report	4-Page Interactive Report

01 BUSINESS CONTEXT

A mid-sized retail company operating across four geographic regions — **East, North, South, and West** — sells products across three major categories: **Technology, Furniture, and Office Supplies**. The company serves three distinct customer segments: **Consumer, Corporate, and Home Office**.

Despite recording strong gross revenue of **\$4.96M** across 2023–2024, the company's leadership team is concerned about stagnating profitability, regional performance inconsistencies, and growing operational costs. The finance and operations teams have raised concerns that sales growth is not translating into proportional profit growth — a critical signal that the business model needs a data-driven review.

02 CORE BUSINESS PROBLEM

"The company is generating significant revenue but cannot explain WHY profit margins remain at only 10% — and WHY profit declined in 2024 despite stable or growing sales. Leadership lacks a unified, data-driven view to identify which regions, categories, segments, and products are hurting overall profitability."

The following specific pain points were identified:

P1	Profit margin stuck at 10% Revenue is growing but profit is not scaling proportionally, indicating margin leakage through discounting, poor product mix, or operational inefficiency.
P2	Profit declined in 2024 vs 2023 Despite stable sales, total profit dropped in 2024 — pointing to increased costs, aggressive discounting, or a shift toward lower-margin products.
P3	Regional performance gap Approval and profit rates vary significantly by region with no clear understanding of root causes — the North outperforms the West in profit but no action has been taken.

P4	No unified category-level insight Management cannot identify which product categories are driving vs. dragging profitability at the region-segment intersection.
P5	Low-margin products undetected Products like Keyboards, Chairs, and Desks are generating high sales volume but low profitability — draining resources without proportional return.

03 KEY METRICS AT A GLANCE

\$4.96M	\$4,97,610	10%	2,000	15.1%	4.0 Days
Total Sales	Total Profit	Profit Margin	Total Orders	Avg Discount	Avg Ship Days

04 ANALYTICAL APPROACH & METHODOLOGY

Step 1	Data Collection & SQL Cleaning	Raw retail data ingestion, handling missing values, fixing NULLs, and standardizing column formats (Product ID, Region, Date).
Step 2	Power BI Integration via Live Database	The cleaned data is connected to Power BI using a live database connection (Get Data > Database > SQL Server).
Step 3	Data Modeling & DAX Measures	A star-schema model is built with Sales, Profit, and Region as fact tables, and Product, Category, and Time as dimension tables for identification and analysis.
Step 4	4-Page Interactive Report Design	The final report is designed for executive decision-making, featuring Analysis (trends, regional performance) and Recommendations (pricing, inventory).

05 KEY INSIGHTS DISCOVERED

01	Profit declined in 2024 despite stable sales, indicating revenue growth is not converting into margin improvement — likely driven by discounting pressure or product mix shift.
02	The North region leads profit generation at \$133K, while West trails at \$119K — a \$14K gap that represents a strategic opportunity for regional pricing review.

03	Month-to-month profit fluctuations show no consistent growth pattern, revealing operational inconsistency in margin management throughout the year.
04	South region leads profit margin at 10.21%, while West lags at 9.92% — with Furniture being the key differentiator across all regions.
05	Consumer, Corporate, and Home Office segments contribute nearly equally to profit (~\$162K–\$168K), suggesting no single segment is dominant — diversification is a strength but also a risk.
06	Top 5 low-profit products (Keyboard, Chair, Folder, Desk, Bookshelf) generate high order volumes but compress margins — prime candidates for pricing strategy review.

06 BUSINESS RECOMMENDATIONS

R1	Reduce Discount Dependency	Finding: Pro products and
R2	Prioritize High-Margin Segments	Finding: Fo combination
R3	Audit Low-Profit Products	Finding: Ke contribution.

07 TOOLS, SKILLS & TECH STACK

MySQL	Data Cleaning, Deduplication, NULL handling, Column Engineering, Data Type fixing, Query-based transformation
Power BI Desktop	Data Modeling, Star Schema, Relationships, Report Design, Multi-page interactive layout, Page navigation
DAX	15+ measures including Profit Margin %, YoY comparison, Segment/Region filters, Low-margin product ranking
Data Storytelling	4-page analytical narrative: Problem → Analysis → Insight → Recommendation → Action Plan
MySQL → Power BI Pipeline	Live database connection using Get Data > MySQL Server method — simulating real enterprise BI workflow